Institute of Electrical and Electronics engineering

Implementation of a PLC based Automated Bottle Cap and Label Inspection System with Machine Learning

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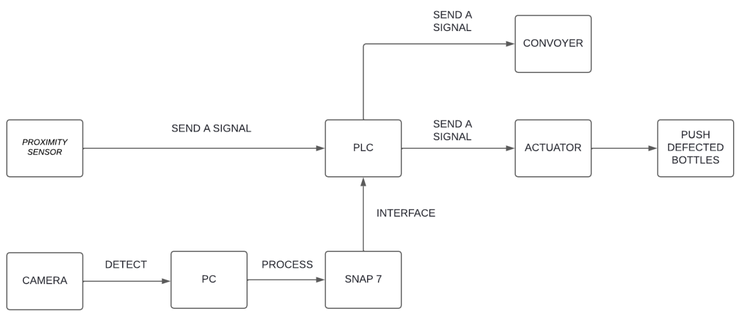
ABSTRACT

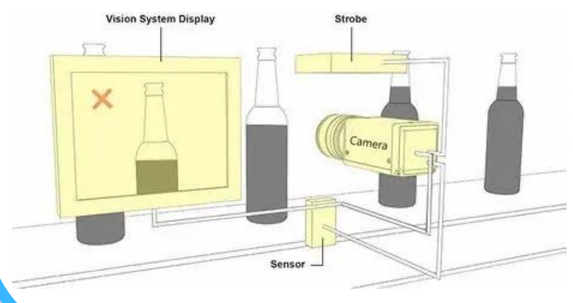
The demand for high-quality control standards in manufacturing industries has led to the development of advanced automated inspection systems. This project presents an innovative approach for bottle inspection utilizing the YOLO (You Only Look Once) deep learning model.YOLO is a state-of-the-art, real-time object detection system known for its high speed and accuracy, making it ideal for optimizing industrial performance.

OVERVEIW

This project aims to create an automated system for identifying and ejecting defective bottles on a conveyor belt. It involves developing software to inspect bottles by capturing frames using computer vision.

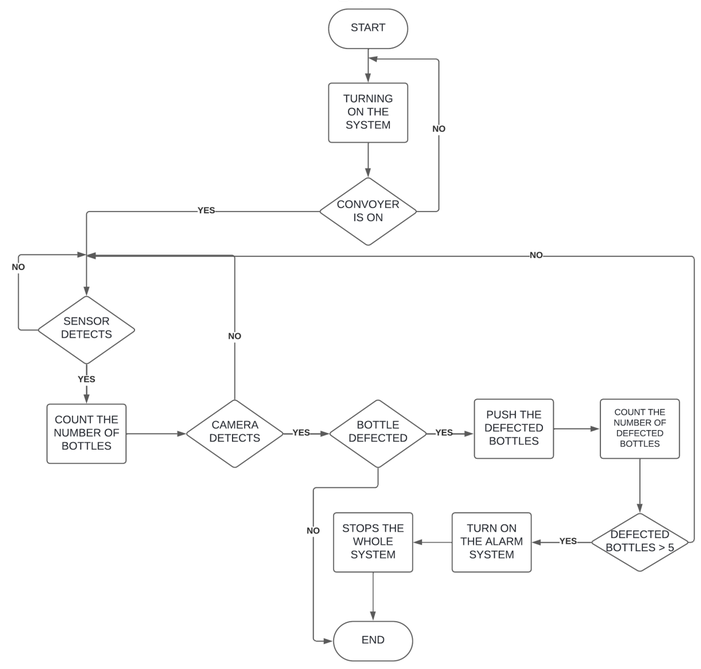


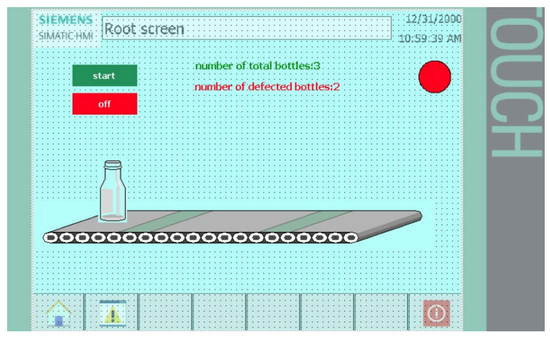
The system uses a trained YOLO model to analyze images and flag defects, then sends a digital signal to a PLC via the SNAP 7 library. The second part involves designing Tia-Portal ladder logic to manipulate the PLC based on signals from proximity sensors and the computer vision program. This controls the actuator and LED alarm, which activates after detecting more than 5 defective bottles.



APPLICATION

The Flow chart below describes the system sequence of processes ,The main logic of the program relies on the condition function, where PLC makes decisions according to the inputs value .Moreover we had developed an HMI to make the controlling process easier .





CONCLUSION

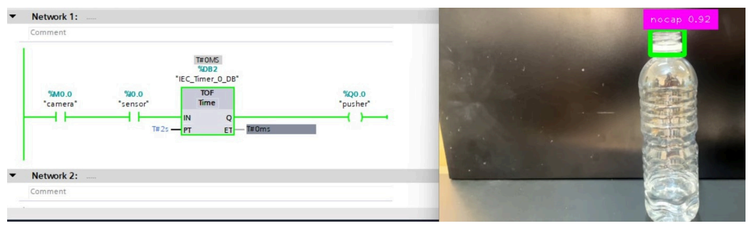
Bottle defects detection is of paramount importance in the bottling process . Bottles are used for different purposes in different industries and markets. Water bottling companies are an example of those industries , As water is very important for humans, its quality is crucial to for this purpose we developed an optimized system that will improve the quality control process within manufacturing , The YOLO model's ability to perform real-time object detection with high accuracy and speed makes it an ideal solution for inspecting bottles .

Future work will aim to refine the model further, enhance its adaptability to different types of defects and products, and integrate additional features such as predictive maintenance and analytics.

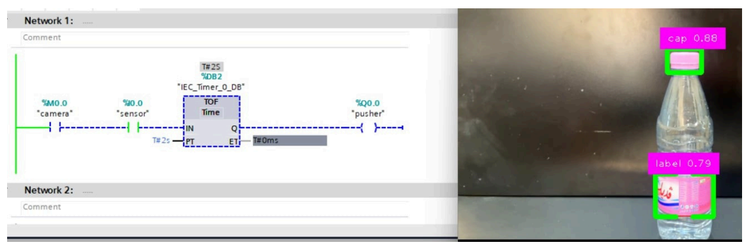
Overall we can say that this project was successfully implemented ,bringing significant benefits to industrial world ,especially production process by enhancing its efficiency and reducing costs.

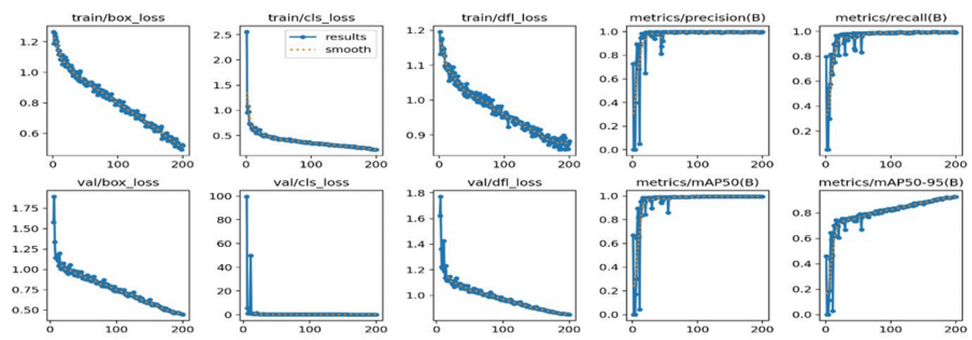
RESULTS

After finishing the implementation and confirming that all our components are placed correctly we had tested some bottles in good lighting conditions , so we got two different results . One shows the result of a good bottle we can see that the pusher is OFF, second displays the result of a defected bottle which will make the actuator ON to push it away the results are shown below



The network achieved excellent results, after few epochs the mAP at 0.5 tends to one ,and the mAP at 0.5:0.95 reaches approximately 0.8 and the losses waveforms are perfect as shown in the following figure

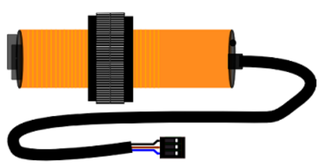




MATERIALS

1.Linear Actoator

2.Proximity Sensor



3.PLC S1200

4.Google Collab

5.TIA Portal

6.VS Code